Artificial Intelligence

Course Information

- **Course Code:** CSC462 Course
- **Title:** Artificial Intelligence
- **Credit Hours:** 4 (3,1)

Text and Reference Books

- **Textbook:** 1. Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.
- **Reference Book:** 1. Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.

Artificial Intelligence Search Agents Informed search

Topics Covered: Heuristic Function, Greedy Best First Search, and A* search.

Reading Material: AIMA: Ch3

Informed search

- In **Artificial Intelligence**, an **informed search algorithm** uses **heuristic information** to guide the search towards the goal **more efficiently** than an **uninformed (blind) search.**
  This extra knowledge helps reduce **search time**, **memory usage**, and **irrelevant exploration**.

Heuristic Function

- A **heuristic function**, often written as **h(n),** is a **guide** that helps an AI algorithm **estimate how close a state is to the goal**.
  It provides ***extra knowledge*** about the problem to **improve search efficiency**.
- The value of the heuristic function is always **positive.**

Think of it like **Google Maps**:

- g(n) = Distance you **have already traveled**.
- h(n) = **Estimated distance left** to reach your destination.
- f(n) = Total estimated cost → **f(n) = g(n) + h(n).**

Heuristic Function

Purpose of Heuristic Function

- **Guides the search** towards the most promising paths.
- **Reduces time and memory usage** by avoiding unnecessary exploration.
- Helps **informed search algorithms** like A*, Greedy Best-First, etc., work efficiently.
- Can **determine optimality** : if the heuristic is accurate, the algorithm can find the best solution.

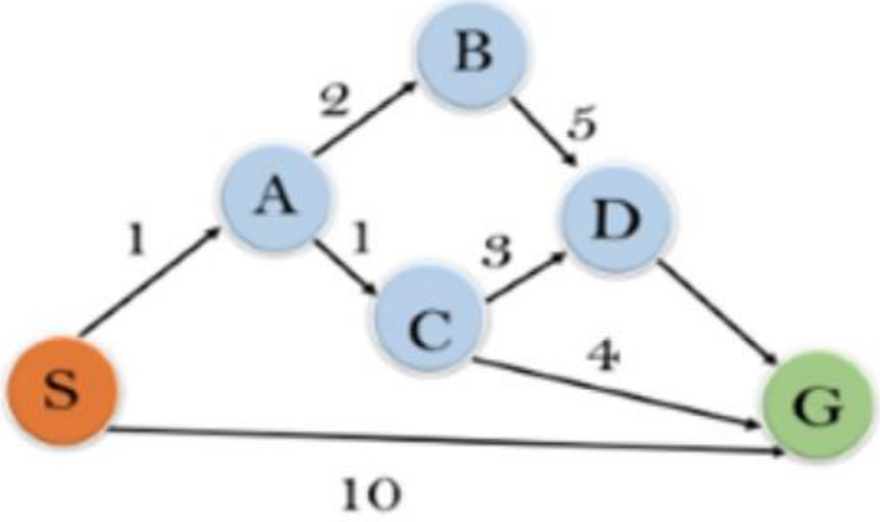
*A * Search Algorithm*

- The A * algorithmis a popular pathfinding and graph traversal algorithm used to find the **shortest path between two nodes**.
- uses a **cost function** to evaluate nodes and prioritizes the most promising paths.
- **Cost Function:**
- A* uses a cost function f(n) to evaluate nodes:
- **f(n)=g(n)+h(n)**
- **g(n):** The actual cost to reach the current node n from the start node.
- **h(n):** The heuristic estimate of the cost to reach the goal node from node n.
- **f(n):** The total estimated cost of the path through node n.

Where A* Can Be Used

Search Type	Used?
Tree Search	Yes
Graph Search	Yes (with a closed list)
Directed Graphs	Yes
Weighted Graphs	Yes
Negative Weights	No

Example 1:



State	h(n)
S	5
A	3
B	4
C	2
D	6
G	0

$f(n) = g(n) + h(n)$

Step 1: Initialization

$S \quad h(S) = 5$

iteration 1: $S \longrightarrow A =$

$f(n) = g(n) + h(n)$

$f(n) = 1 + 3 = \boxed{4}$ ✓

$S \rightarrow G$

$f(n) = g(n) + h(n)$

$f(n) = 10 + 0 = \boxed{10}$

iteration 2: $S \rightarrow A \rightarrow B$

$f(n) = g(n) + h(n)$

$f(n) = (1+2) + 4$

$\boxed{f(n) = 7}$

$S \rightarrow A \rightarrow C$

$f(n) = g(n) + h(n)$

$f(n) = (1+1) + 2$

$f(n) = \boxed{4}$ ✓

$S \rightarrow G$

$10 + 0 = \boxed{10}$

iteration 3:

$S \rightarrow A \rightarrow C \rightarrow D$

$f(n) = g(n) + h(n)$

$f(n) = (1+1+3) + 6 = \boxed{11}$

$S \rightarrow A \rightarrow B = \quad f(n) = g(n) + h(n)$

$f(n) = (1+2) + (4) = \boxed{7}$

$S \rightarrow G = \boxed{10}$

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$S \to A \to C \to G$

$f(n) = g(n) + h(n)$
$f(n) = (1 + 1 + 4) + 0 = 6$

$\boxed{f(n) = 6}$

<u>iteration 4</u> :

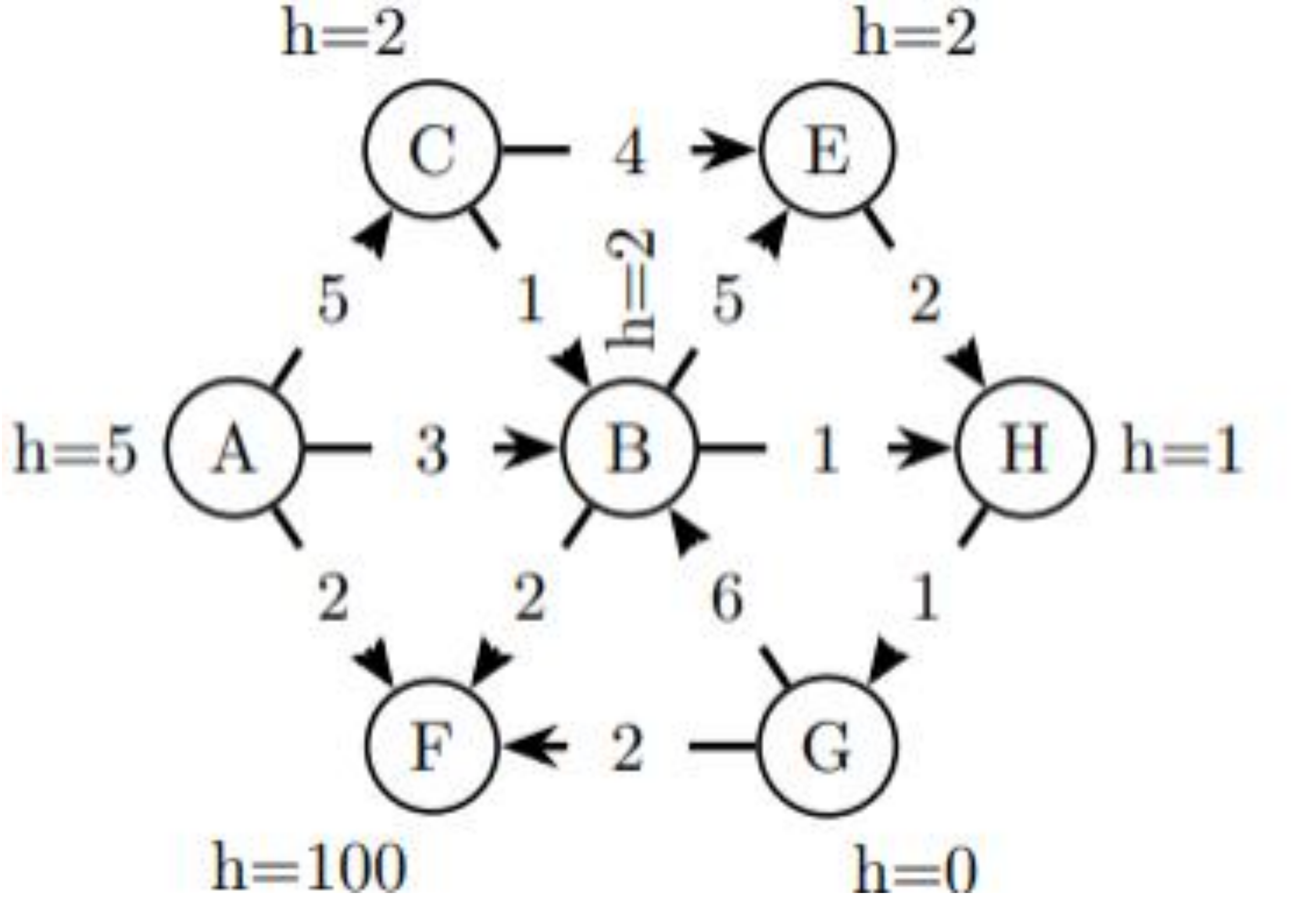
We will give the final result as

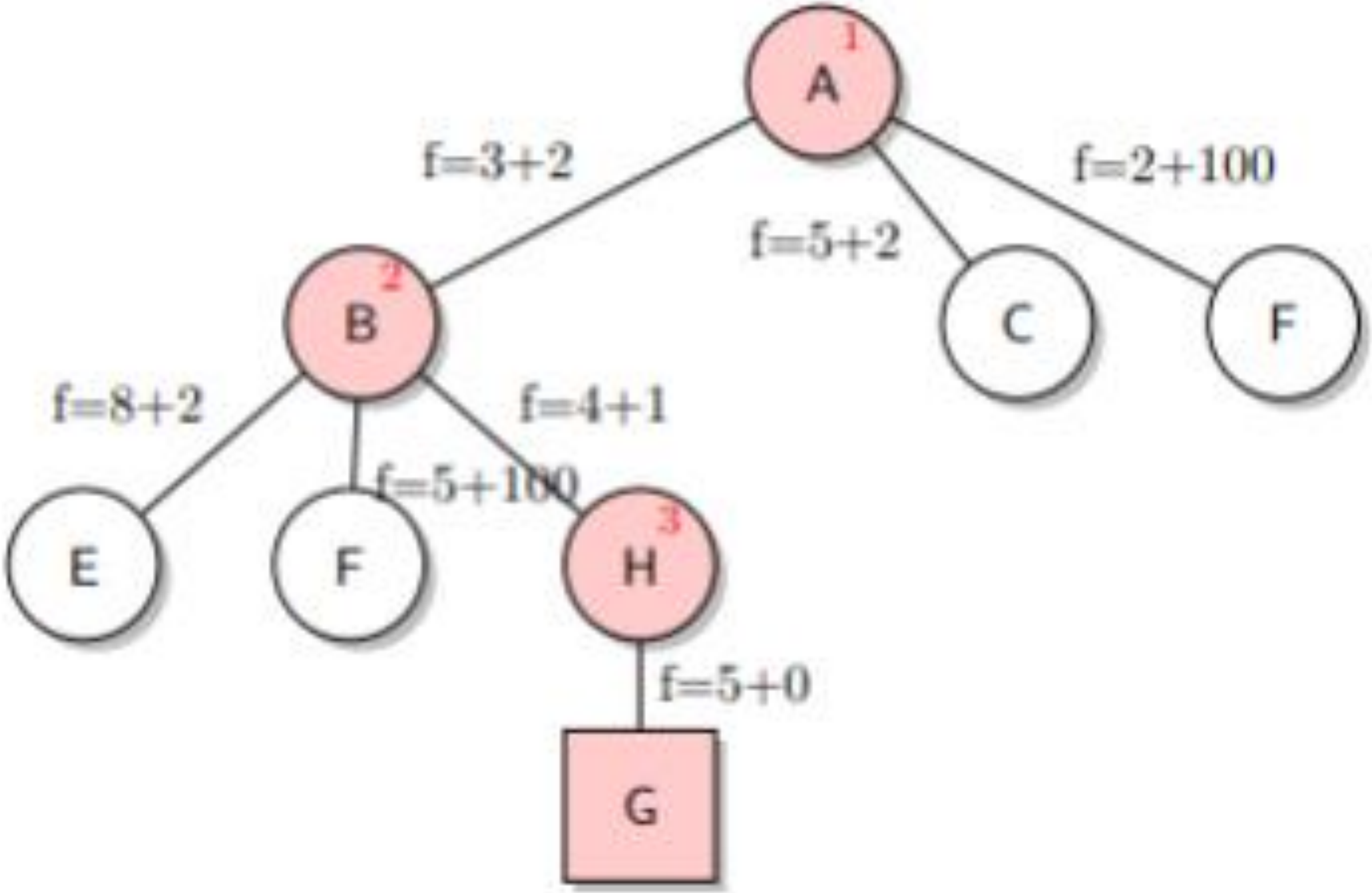
$S \to A \to C \to G$

it provide the optimal path with Cost $\boxed{6}$

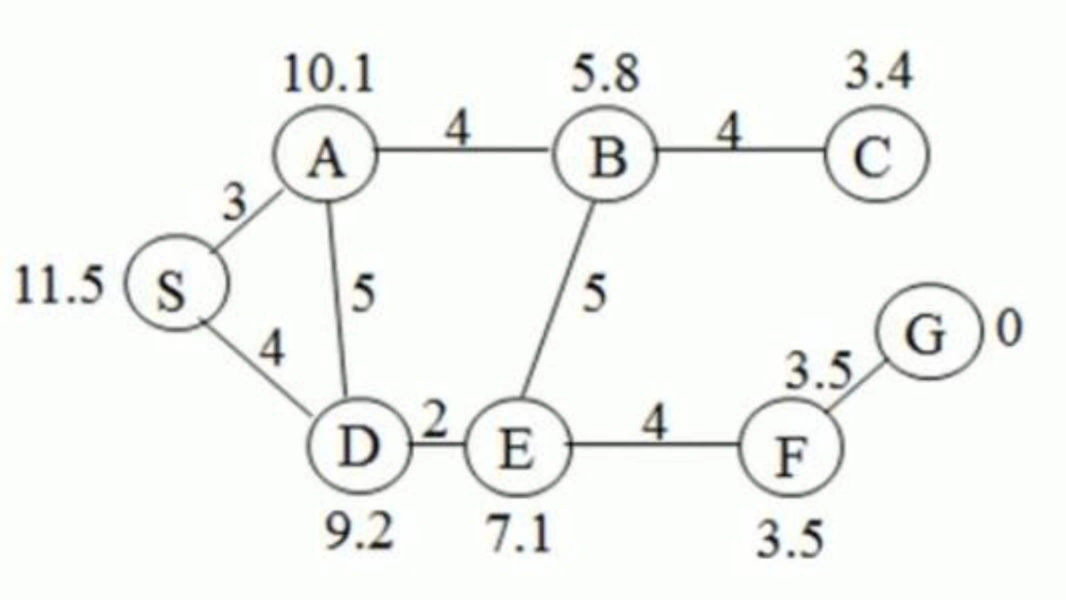
Example 2

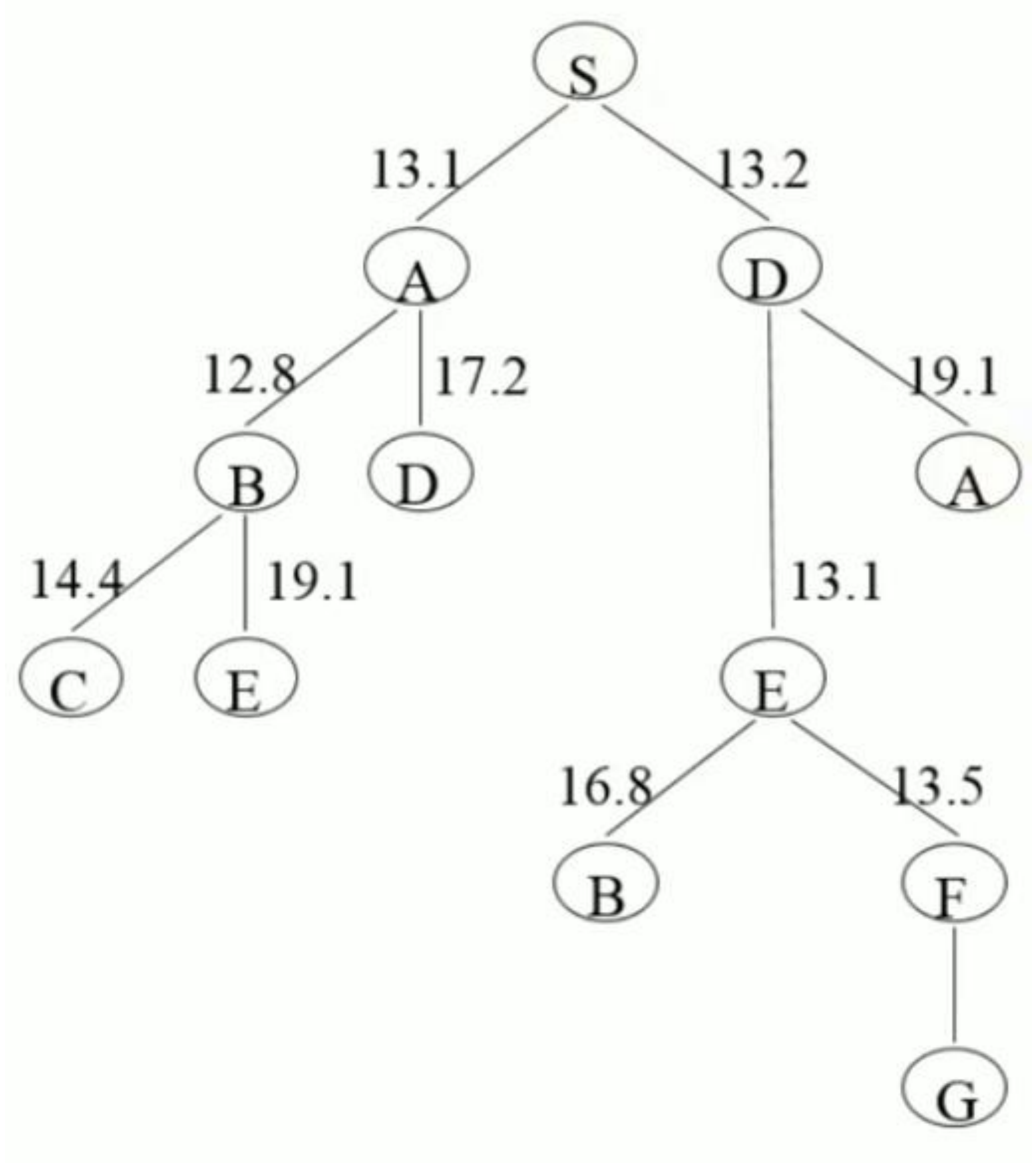
- Given the search problem defined by the following graph, **A** is the initial state and **G** is the goal state and the cost of each action is indicated on the corresponding edge.



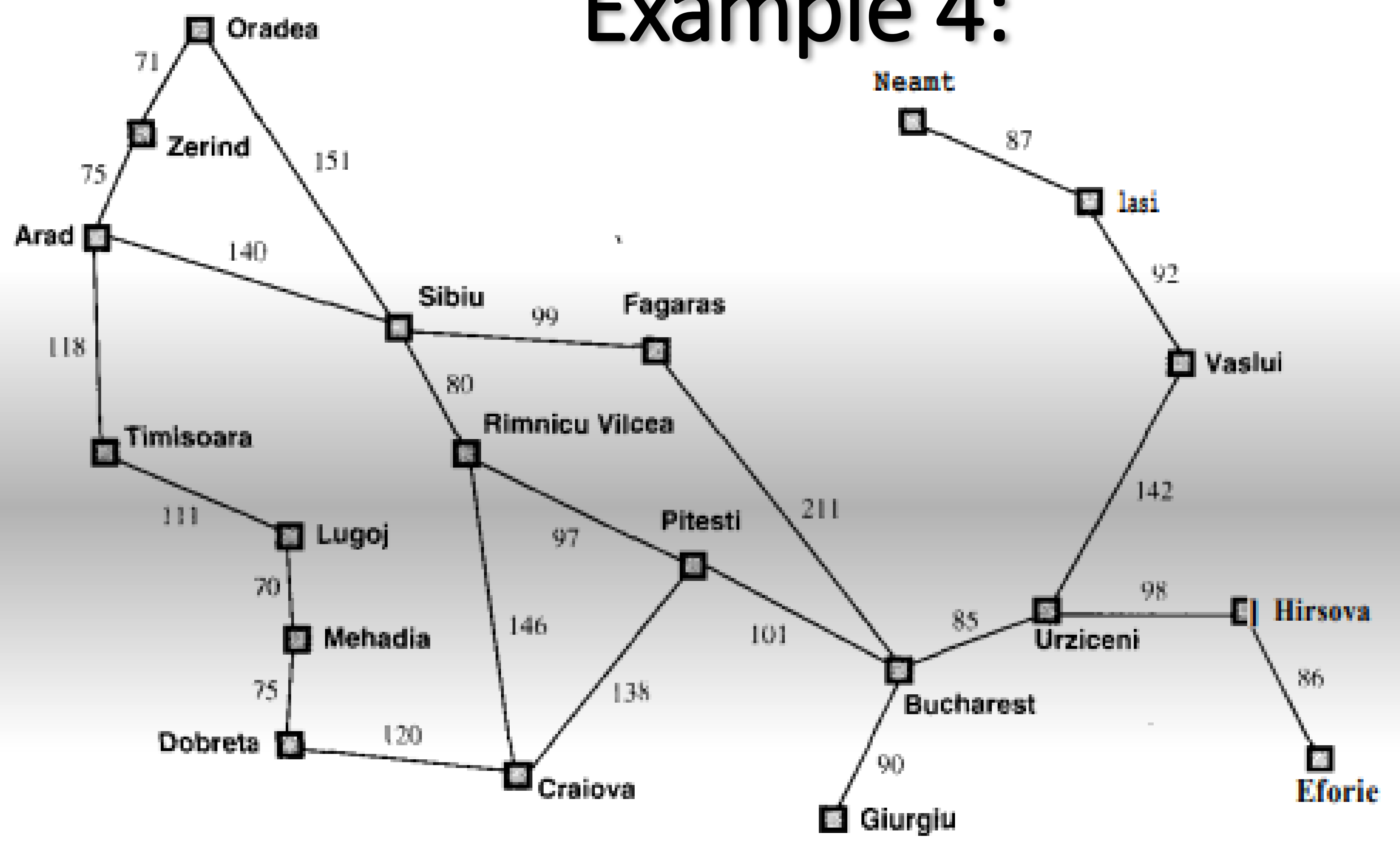

A
1
f=3+2
f=5+2
f=2+100
B
2
C
F
f=8+2
f=4+1
f=5+100
E
F
H
3
f=5+0
G

Example 3:




S
13.1
13.2
A
D
12.8
17.2
19.1
B
D
A
14.4
19.1
13.1
C
E
E
16.8
13.5
B
F
G

Example 4:



Straight-line distance to Bucharest

Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Figure 4.2 Map of Romania with road distances in km, and straight-line distances to Bucharest.

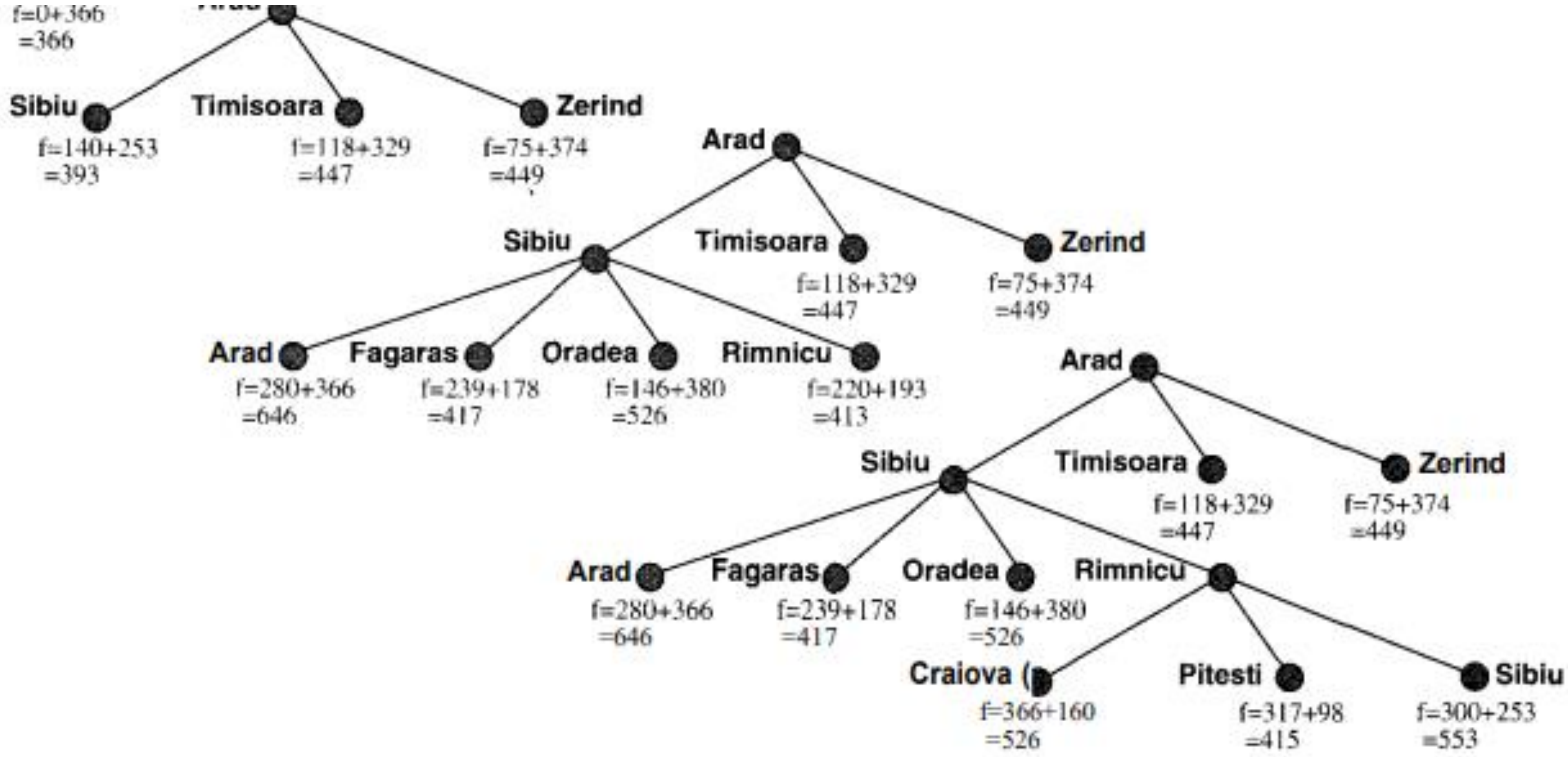


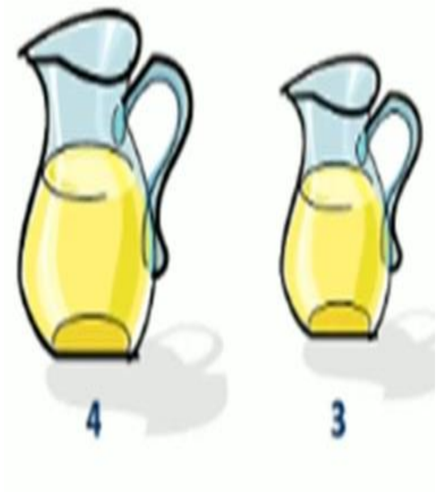
Figure 4.4 Stages in an A* search for Bucharest. Nodes are labelled with / = g + *h*. The *h* values are the straight-line distances to Bucharest taken from Figure 4.1.

Water Jug Problem in Artificial Intelligence

Water Jug Problem in Artificial Intelligence

Water Jug Problem Definition,

- You are given two jugs, a 4-liter one and a 3-liter one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 liters of water into a 4-liter jug.



Representation of water Jug Problem in terms of state-space search,

- State: (x, y)
- where x represents the quantity of water in a 4-liter jug and y represents the quantity of water in a 3-liter jug.
- That is, x = 0, 1, 2, 3, or 4 y = 0, 1, 2, 3
- Start state: (0, 0)
- Goal state: (2, n) for any n.
- Here we need to start from the start state and end up in a goal state.

1	(x, y) is X<4 ->(4, Y)	Fill the 4-liter jug
2	(x, y) if Y<3 -> (x, 3)	Fill the 3-liter jug
3	(x, y) if x>0 -> (x-d, y)	Pour some water out of the 4-liter jug.
4	(x, y) if Y>0 -> (x, y-d)	Pour some water out of the 3-liter jug.
5	(x, y) if x>0 -> (0, y)	Empty the 4-liter jug on the ground
6	(x, y) if y>0 -> (x,0)	Empty the 3-liter jug on the ground
7	(x, y) if X+Y >= 4 and y>0 -> (4, y-(4-x))	Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full
8	(x, y) if X+Y>=3 and x>0 -> (x-(3-y), 3))	Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.
9	(x, y) if X+Y <=4 and y>0 -> (x+y, 0)	Pour all the water from the 3-liter jug into the 4-liter jug.
10	(x, y) if X+Y<=3 and x>0 -> (0, x+y)	Pour all the water from the 4-liter jug into the 3-liter jug.
11	(0, 2) -> (2, 0)	Pour the 2-liter water from the 3-liter jug into the 4-liter jug.
12	(2, Y) -> (0, y)	Empty the 2-liter in the 4-liter jug on the ground.

The solution to Water Jug Problem in Artificial Intelligence

1. Start state = (0, 0)

2. Loop until the goal state (2, 0) reached

   - Apply a rule whose left side matches the current state

   - Set the new current state to be the resulting state

1. Solution to Water Jug Problem in Artificial Intelligence

- (0, 0) – Start State
- (0, 3) – Rule 2, Fill the 3-liter jug
- (3, 0) – Rule 9, Pour all the water from the 3-liter jug into the 4-liter jug.
- (3, 3) – Rule 2, Fill the 3-liter jug
- (4, 2) – Rule 7, Pour water from the 3-liter jug into the 4-liter jug until the 4-liter jug is full.
- (0, 2) – Rule 5, Empty the 4-liter jug on the ground
- (2, 0) – Rule 9, Pour all the water from the 3-liter jug into the 4-liter jug.
- Goal State reached

2. Solution to Water Jug Problem in Artificial Intelligence

- (0, 0) – Start State
- (4, 0) – Rule 1, Fill the 4-liter jug
- (1, 3) – Rule 8, Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.
- (1, 0) – Rule 6, Empty the 3-liter jug on the ground
- (0, 1) – Rule 10, Pour all the water from the 4-liter jug into the 3-liter jug.
- (4, 1) – Rule 1, Fill the 4-liter jug
- (2, 3) – Rule 8, Pour water from the 4-liter jug into the 3-liter jug until the 3-liter jug is full.